Journal of Clinical Epidemiology 181 (2025) 111746

Journal of Clinical Epidemiology

ORIGINAL RESEARCH

Large language models for conducting systematic reviews: on the rise, but not yet ready for use—a scoping review

Judith-Lisa Lieberum[a], Markus Toews[b], Maria-Inti Metzendorf[c], Felix Heilmeyer[d], Waldemar Siemens[b], Christian Haverkamp[d], Daniel Böhringer[a], Joerg J. Meerpohl[b,e,1], Angelika Eisele-Metzger[b,e,*,1]

[a]*Eye Clinic, Medical Center − University of Freiburg/Medical Faculty − University of Freiburg, Freiburg, Germany*
[b]*Institute for Evidence in Medicine, Medical Center − University of Freiburg/Medical Faculty − University of Freiburg, Freiburg, Germany*
[c]*Institute of General Practice, Medical Faculty of the Heinrich-Heine-University Düsseldorf, Düsseldorf, Germany*
[d]*Institute for Digitalization in Medicine, Medical Center − University of Freiburg/Medical Faculty − University of Freiburg, Freiburg, Germany*
[e]*Cochrane Germany, Cochrane Germany Foundation, Freiburg, Germany*

Accepted 20 February 2025; Published online 26 February 2025

Abstract

Background and Objectives: Machine learning promises versatile help in the creation of systematic reviews (SRs). Recently, further developments in the form of large language models (LLMs) and their application in SR conduct attracted attention. We aimed at providing an overview of LLM applications in SR conduct in health research.

Methods: We systematically searched MEDLINE, Web of Science, IEEEXplore, ACM Digital Library, Europe PMC (preprints), Google Scholar, and conducted an additional hand search (last search: February 26, 2024). We included scientific articles in English or German, published from April 2021 onwards, building upon the results of a mapping review that has not yet identified LLM applications to support SRs. Two reviewers independently screened studies for eligibility; after piloting, 1 reviewer extracted data, checked by another.

Results: Our database search yielded 8054 hits, and we identified 33 articles from our hand search. We finally included 37 articles on LLM support. LLM approaches covered 10 of 13 defined SR steps, most frequently literature search (n = 15, 41%), study selection (n = 14, 38%), and data extraction (n = 11, 30%). The mostly recurring LLM was Generative Pretrained Transformer (GPT) (n = 33, 89%). Validation studies were predominant (n = 21, 57%). In half of the studies, authors evaluated LLM use as promising (n = 20, 54%), one-quarter as neutral (n = 9, 24%) and one-fifth as nonpromising (n = 8, 22%).

Conclusion: Although LLMs show promise in supporting SR creation, fully established or validated applications are often lacking. The rapid increase in research on LLMs for evidence synthesis production highlights their growing relevance. © 2025 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Keywords: Machine learning; Large language models; ChatGPT; Systematic reviews as topic; Scoping review; Health research

Funding: This work was supported by the Research Commission at the Faculty of Medicine, University of Freiburg, Freiburg, Germany (grant no. EIS2244/23). Open Access funding enabled and organized by Project DEAL.

[1] Equally contributed.

* Corresponding author. Medical Faculty− University of Freiburg, Institute for Evidence in Medicine, Medical Center − University of Freiburg, Breisacher Straße 86, Freiburg im Breisgau 79110, Germany.
E-mail address: angelika.eisele-metzger@uniklinik-freiburg.de (A. Eisele-Metzger).

https://doi.org/10.1016/j.jclinepi.2025.111746
0895-4356/© 2025 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Plain Language Summary

Systematic reviews are a crucial tool in health research where experts carefully collect and analyze all available evidence on a specific research question. Creating these reviews is typically time- and resource-intensive, often taking months or even years to complete, as researchers must thoroughly search, evaluate, and synthesize an immense number of scientific studies. For the present article, we conducted a review to understand how new artificial intelligence (AI) tools, specifically large language models (LLMs) like Generative Pretrained Transformer (GPT), can be used to help create systematic reviews in health research. We searched multiple scientific databases and finally found 37 relevant articles. We found that LLMs have been tested to help with various parts of the systematic review process, particularly in 3 main areas: searching scientific literature (41% of studies), selecting relevant studies (38%), and extracting important information from these studies (30%). GPT was the most commonly used LLM, appearing in 89% of the studies. Most of the research (57%) focused on testing whether these AI tools actually work as intended in this context of systematic review production. The results were mixed: about half of the studies found LLMs promising, a quarter were neutral, and one-fifth found them not promising. While LLMs show potential for making the systematic review process more efficient, there is still a lack of fully tested and validated applications. However, the increasing number of studies in this field suggests that these AI tools are becoming increasingly important in creating systematic reviews.

1. Introduction

Systematic reviews (SRs) form the basis for evidence-based medicine [1,2] but are time- and resource-intensive, requiring several researchers [3]. Using artificial intelligence (AI) to assist with the different SR tasks offers a promising approach to save time and decrease personnel demands [4,5].

Recently, a wide array of AI tools has evolved. Traditional machine learning (ML) relies on supervised or unsupervised algorithms for task-specific decisions [6–9]. Transformers like Bidirectional Encoder Representations from Transformers (BERT) [10] vastly improved semantical and contextual language processing. Generative large language models (LLMs) such as Generative Pretrained Transformer (GPT) [11], Large Language Model Meta AI (LLaMA) [12] or Claude [13] follow instructions in natural language without task-specific training [14]. These are built on decoder-only transformers and trained on vast amounts of textual data. Yet, their opaque architecture raises risks like harmful responses or misinformation [14–16]. Currently, LLMs are extensively tested in medicine [17] and health research [18].

Several ML-based tools already assist SR conduction in health research, including ASReview for screening [19] and DistillerSR for various SR steps [20]. In a mapping review based on an April-May 2021 systematic search, Cierco Jimenez et al [4] identified a broad range of ML tools that assist in SR performance. However, no LLM applications—perhaps holding even greater promise—could be identified at that time. Since then, LLM use in this context has risen significantly, aiding in tasks like formulating review questions [21], screening [22], or data extraction [23]. However, these approaches are still experimental and error prone. For instance, our recent attempts to assess risk of bias (RoB) with Cochrane's risk of bias tool "RoB2" using Claude achieved limited success, far from replacing human reviewers [24].

This scoping review aims to provide an overview of recent approaches to facilitate SR conduct with the help of LLMs, highlighting the most promising strategies and forming a basis for future advancement and critical evaluation.

2. Methods

This scoping review was conducted based on the guidelines of the Joanna Briggs Institute [25] and reported in line with the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) extension for Scoping Reviews statement [26].

2.1. Protocol and registration

We registered the protocol for this scoping review on Open Science Framework (OSF) on March 4, 2024 (https://osf.io/asjm3). Changes from the protocol are listed in the supplement.

2.2. Eligibility criteria

Eligibility criteria were defined following the population, concept, context framework [25]; as common for methodological scoping reviews; limitations regarding population were not applicable. We included articles on LLM applications (concept) in the context of conducting an SR in health research, published from April 1, 2021 onwards, building on the results of an earlier review on ML applications by Jimenez et al who conducted their search in April and May 2021 and had not yet identified any LLM applications to support SRs [4]. Due to feasibility, inclusion was restricted to full scientific articles in English and German language, published

What is new?

Key findings

- We found 37 articles describing LLM support for SR production. Many of their conclusions were positive.

What this adds to what was known?

- Research on LLM support for SR conduct is on the rise. However, many approaches are currently exploratory in nature.

What is the implication and what should change now?

- LLMs should only be used with caution and under human supervision to support the SR process. Methodologically sound validation studies are highly desirable and necessary before a broad use of LLMs is possible in this field.

in journals or on preprint servers. Preprint articles were included to capture the most up-to-date developments in this rapidly evolving field. We considered support of any individual step or of the entire SR process. Specification of SR steps considered are listed in our protocol.

Originally, we had planned to also include and describe all applications of ''traditional'' ML (as outlined in our protocol). Due to the large number of included articles on ML approaches as well as the growing awareness of the importance of the emerging LLM technology for SR support in the course of our work, we decided to focus this report on LLM use. References and extracted data of the included articles on more traditional ML were uploaded to OSF (https://osf.io/vdsgb) and are briefly summarized in the supplement.

We excluded study or review protocols, preclinical literature, sources describing the application of tools specifically for guideline development and sources lacking detail regarding the tools and methods used or the area of application. Relevant systematic, scoping or mapping reviews as well as surveys or guidance articles mentioning a range of approaches were included, but used for citation searching only, that is, to check whether they cite further relevant articles that we may have missed with our systematic search.

2.3. Information sources

We systematically searched MEDLINE via Ovid, Web of Science Core Collection (Science Citation Index Expanded, Social Science Citation Index, Emerging Sources Citation Index), IEEEXplore and ACM Digital Library (journals only), Europe PMC (preprints only) and Google Scholar (first 200 hits, sorted by relevance).

We conducted backward citation searching considering the relevant review articles identified by our search as well as the review by Parisi and Sutton [27] on the role of ChatGPT in systematic literature searches published during our SR process. Additionally, we searched the Digital Evidence Synthesis Tool Evaluations [28] and a range of literature collected from preliminary searches.

2.4. Search

We established the search strategy based on text analysis of 38 known relevant records. Of those, 24 were indexed in PubMed, on which the search strategy for MEDLINE was developed by analyzing the most frequent Medical Subject Headings (MeSH) terms with Yale MeSH Analyzer and conducting text word frequency analyses with PubReMiner. Additionally, we used parts of a search filter for generative AI [29]. We subsequently revised the search strategy to capture the 14 records that were not indexed in PubMed and adapted the final search strategy to all sources. The date of search for all databases was February 26, 2024. The search results were deduplicated using Deduklick [30], resulting in 8054 records from database searches to be screened. Our final search strategies can be found in the supplement. In September 2024, we reviewed all preprint articles for a possible change of status to a peer-reviewed article, and used the journal article for data charting, if available.

2.5. Selection of sources of evidence

The review team for source selection consisted of three reviewers (J.L.L., A.E.M., and M.T.), all of who completed the initial pilot testing of 25 randomly selected sources: Using Covidence review software [30], two out of three reviewers independently reviewed eligibility of all records first on title and abstract basis, then as full-text screening for the remaining articles. At each stage, we resolved discrepancies by discussion.

2.6. Data charting process and data items

We developed a customized Covidence spreadsheet. We conducted a pilot testing with three randomly chosen sources extracted by two reviewers independently. After finalizing the spreadsheet and obtaining sufficient agreement during piloting, LLM data were extracted by 1 reviewer (J.L.L.) and verified by a second (A.E.M.). We extracted the types of LLM(s) used, the SR step(s) supported, location of the primary author, type of article, study design, methods in brief, key findings, limitations, funding, and conflicts of interest. We collected authors' overall conclusions and rated them as ''promising'', ''neutral'', or ''non-promising''. The rating was made jointly by two reviewers (J.L.L. and A.E.M.) in a content analytical manner. It was primarily based on the statements of the study authors in the discussion and conclusions sections of their articles. Thus, the rating reflects the study authors' own appraisal

and interpretation of their results. Additionally, we considered the study results and performance metrics such as accuracy, precision, and recall. For transparency, we collected supporting quotes and, if available, performance metrics for each rating to underpin our decision. These data are included in the supplement. For study design, we categorized studies as ''validation study'' if a defined reference standard (or a human expert check) was used and matching to this standard was calculated.

Due to the nature of this review (scoping review), we did not critically appraise the underlying sources.

2.7. Synthesis of results

We converted extracted data from Covidence to Microsoft Excel 2016, performed statistical evaluations of frequencies with Excel and Stata (version 16.1), and generated graphical charts with R (version 4.4.2) and Microsoft PowerPoint 2016. Detailed data tables were uploaded to OSF (https://osf.io/vdsgb).

3. Results

Our systematic search of 6 databases yielded 11,323 hits from databases and further 33 records identified via others methods. Details of the selection process are depicted in the PRISMA flowchart (Fig 1). Finally, 196 studies were included, of which 159 focused on more traditional ML applications and 37 on LLM use. All articles included were published in English language. References and extracted data of the included articles on more traditional ML approaches were uploaded to OSF (https://osf.io/vdsgb) and are briefly summarized in the supplement.

In the following, we concentrate on LLMs as the primary focus of our review (overview and characteristics in Table 1; data charting on OSF https://osf.io/vdsgb): The most frequent SR steps supported by LLMs were systematic literature search (n = 15, in 41% of all 37 studies), study selection including title/abstracts and full texts (n = 14, 38%), and data extraction (n = 11, 30%), followed by RoB assessment of primary studies (n = 5, 14%), interpretation of findings (n = 4, 11%), framing the SR question and inclusion criteria (n = 3, 8%), and qualitative/narrative/descriptive summary of findings (n = 3, 8%). LLMs were rarely used for writing PLS (n = 1) and SR publications (n = 2), or quantitative data analysis (n = 2).

In terms of LLMs reported, GPT was the tool used by far most commonly in the studies (n = 33, 89% of studies), followed by LLaMA (n = 3, 8%), and Claude (n = 3, 8%). Details regarding the kind of LLM used for each SR step are depicted in Figure 2 (middle donut) and listed in Table 1, including LLM subclasses.

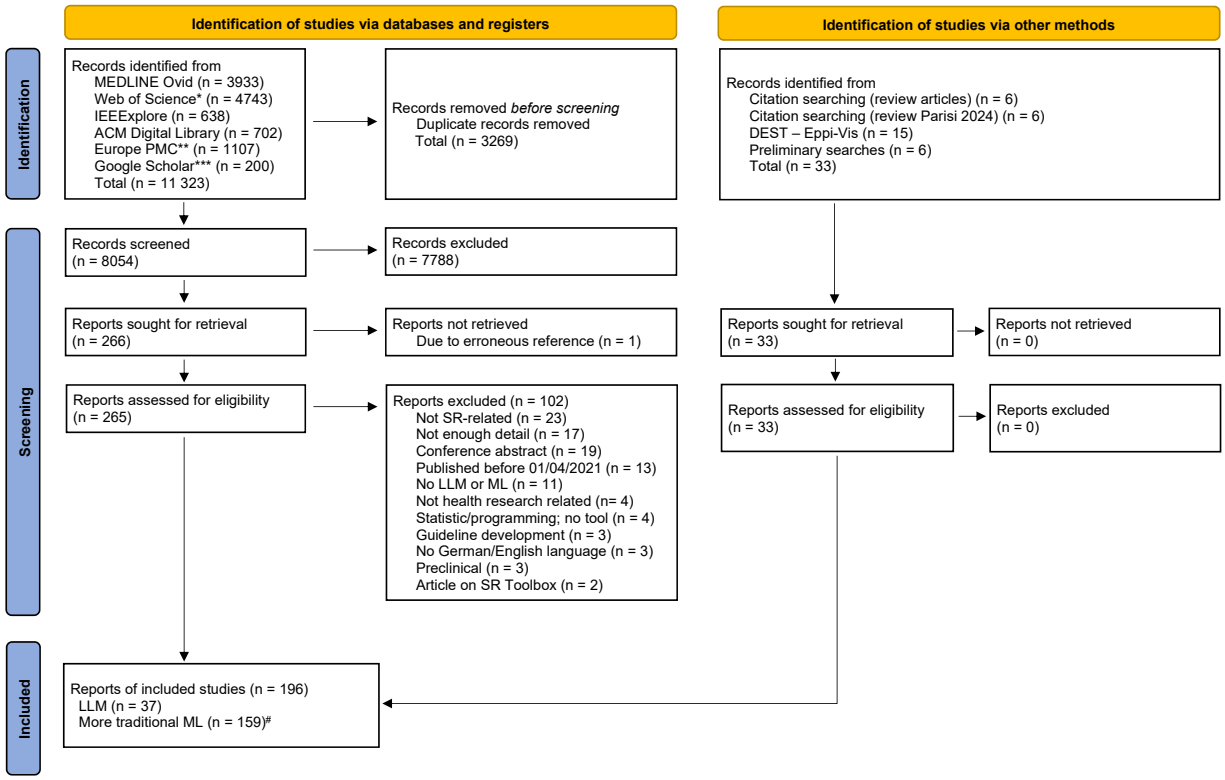


Figure 1. PRISMA flowchart describing the source of evidence retrieval and selection process of reports on the use of large language models (LLMs) and machine learning (ML) in systematic review (SR) conduction in health research. * Web of Science Core Collection, ** Preprints only, *** First 200 hits sorted by relevance, # Results included in the supplement and on OSF (https://osf.io/vdsgb).

Table 1. Overview and characteristics of the included studies with LLM applications to support conducting systematic reviews

ID, reference, article type	Location[d], study design	Kind of LLM	Systematic review task(s)	Main aim/methods
(1) Alaniz 2023[c]	US, exploratory testing	GPT 4	Search	Refinement of MeSH terms, formulation of a PubMed search string, translation of Boolean queries for other databases
(2) Anghelescu 2023[b]	Romania, exploratory testing	GPT[e]	Search	Comparison of human search with a GPT-based chatbot interrogation
(3) Blum 2023[c]	US, exploratory testing	GPT[e]	Search	General and specific literature search in PubMed via instructions given to ChatGPT, compared to a manual PubMed search
(4) Corti 2023[c]	US, exploratory testing	GPT[e]	Search	Counterchecked gray literature search with ChatGPT by asking specific questions, in addition to a systematic human search
(5) Gargari 2024[c]	Iran, validation study	GPT 3.5	Screening	Title/abstract screening by GPT-3.5 (based on eight prompts), according to given inclusion criteria, compared to human researchers, and with a recently published SR as a ''gold standard''
(6) Gartlehner 2023[b]	US, validation study	Claude 2	Data extraction	Data extraction from published studies, compared with human data extraction
(7) Giunti 2024[b]	Ireland, exploratory testing	GPT 3.5	Search, data extraction	Automatic Python script generation and application for mHealth app data extraction from the Google Play Store, evidence identification by analyzing app descriptions, cross-referencing with scientific publications and output files creation
(8) Guimarães 2024[c]	Brazil, exploratory testing	GPT[e]	Search	Search strategy creation based on a PROSPERO record
(9) Guo 2024[b]	Canada, validation study	GPT 3.5, GPT 4	Screening	Creation of a Python script for title-abstract screening by Chat GPT and GPT-4 API
(10) Gupta 2023a[c]	US, validation study	GPT[e]	SR question	Novel SR question creation by ChatGPT with one broad and 2 specific commands
(11) Gupta 2023b[b]	US, validation study	GPT 3.5	SR question	Production of novel SR ideas by ChatGPT with given commands
(12) Haman 2023[c]	Czech Republic, exploratory testing	GPT[e]	Search	Searches for articles in the field of medicine by ChatGPT via prompts, manual countercheck
(13) Hasan 2024[b]	US, validation study	GPT 4, GPT 4 Turbo	RoB	Agreement of GPT-4 vs Cochrane reviewers in RoB judgment of nonrandomized studies from published SRs, using the ROBINS-I tool
(14) Hatami 2023[a]	Iran, tool development	GPT[e]	Search, data extraction	Daily records of literature metadata (based on PICO components) to a repository via a tuned Python engine with natural language processing methods of OpenAI API interference and human supervision
(15) Hill 2023[b]	UK, exploratory testing	GPT[e]	Data extraction	Data extraction from PDF documents using BING AI Chat via Microsoft Edge using commands
(16) Huang 2024[a]	Canada, narrative review	GPT 4	Search, data extraction	Narrative review summarizing potential opportunities in information retrieval (primarily) with ChatGPT

(*Continued*)

Table 1. Continued

ID, reference, article type	Location[d], study design	Kind of LLM	Systematic review task(s)	Main aim/methods
(17) Jiang 2023[a]	UK, validation study	GPT 3.5	Screening	Abstract relevance prioritization using a question-answer framework
(18) Kataoka 2023[a]	Japan, validation study	GPT 3.5, GPT 4	Screening	Development and validation of abstract classification regarding diagnostic test accuracy via GPT
(19) Khraisha 2024[b]	Ireland, validation study	GPT 4	Screening, data extraction	Screening of titles/abstracts, full texts and data extraction from given full texts in English and non-English languages by GPT-4, with criteria given via prompts, refined at each stage based on findings
(20) Mahuli 2023[c]	India, exploratory testing	GPT 3.5	RoB, data extraction	RoB assessment with ChatGPT using ROBINS-I tool, and data extraction by sharing sections of the text and specifying the desired information or data to be extracted by ChatGPT
(21) Matsui 2023[a]	Japan, validation study	GPT 3.5	Screening	Prompt engineering for screening with GPT-3.5, with two published SRs as reference and human refinement
(22) McGowan 2023[b]	US, validation study	GPT 3.5, PaLM 2	Search	Output of references on specific medical questions, including DOIs and full texts, with ChatGPT and Bard 2.0 and in different languages
(23) Nashwan 2023[c]	Qatar, opinion	GPT 4, LaMDA, LLaMA	RoB	Exploring innovative LLM applications in conduction of SRs, with specific focus on quality assessment and RoB evaluation
(24) Ovelman 2024[b]	US, validation study	Claude 2	PLS	Plain language summaries with Claude 2 and specific prompts, focusing on PICO elements and reporting the results of the review
(25) Pitre 2023[a]	Canada, validation study	GPT 4	RoB	RoB-assessment by ChatGPT-4 using RoB 2.0-tool and 3 different prompts, based on recently published Cochrane SRs of medical interventions in randomized trials
(26) Qureshi 2023[c]	US, exploratory testing	GPT 3.5, GPT 4	SR question, search, screening, synthesis (qual.), synthesis (quant.), interpretation	Demonstration and evaluation of various SR methodology related tasks carried out by ChatGPT, in form of a webinar hosted by the developers of PICO Portal (AI-assisted SR platform)
(27) Reason 2024[b]	UK, validation study	GPT 4	Data extraction, synthesis (quant.), interpretation	Data extraction, network-meta-analysis and reporting via LLM prompting, based on four manually conducted case studies
(28) Robinson 2023[a]	UK, tool development	GPT[e], LLaMA, Guanaco	Screening	Comparison of different LLMs and Bio-BERT for abstract screening, with instruction tuning based on a training set of existing Cochrane Reviews
(29) Schopow 2023[b]	Germany, validation study	GPT 3.5, GPT 4	Search, screening, data extraction, synthesis (qual.), interpretation, SR publication	Use of ChatGPT for determining MeSH terms, title/abstract- and full-text screening, limited data extraction, and generating parts of an SR publication
(30) Suppadungsuk 2023[b]	US, validation study	GPT 3.5	Search	Use of ChatGPT to provide Reference citations including DOIs and reference links

(*Continued*)

Table 1. Continued

ID, reference, article type	Location[d], study design	Kind of LLM	Systematic review task(s)	Main aim/methods
(31) Syriani 2024[b]	Canada, validation study	GPT 3.5	Screening	Comparative analysis of the performance of ChatGPT, prompted to adapt to different SR strategies, and machine learning techniques widely used in SR automation
(32) Teperikidis 2023[b]	Greece, study within a SR	GPT[e]	Search, screening, RoB, data extraction, synthesis (qual.), interpretation, SR publication	Use of ChatGPT to generate a search string addressing the review question, titles/abstract screening, data extraction, quality assessment using AMSTAR 2, and writing parts of the publication of an overview of SRs
(33) Tran 2023[b]	France, validation study	GPT 3.5	Screening	Comparison of LLM titles/abstracts screening with human authors, based on five published rapid or SRs
(34) Wang 2023a[c]	Australia, validation study	GPT[e]	Search	Creation of Boolean queries for SR literature search with ChatGPT using different prompts, with two standard test collections of SRs as a basis
(35) Wang 2023b[c]	Australia, validation study	GPT 3.5, Alpaca	Screening	Testing and evaluation of document ranking for screening prioritization with ChatGPT or Alpaca, based on natural language queries generated from given Boolean queries
(36) Wang 2024[a]	Australia, validation study	LLaMA, Llama 2, Alpaca, Guanaco, Falcon	Screening	Testing of automatic document screening with different zero-shot LLMs, compared to a baseline model (BioBERT), using available data
(37) White 2023[b]	Peru, study within a SR	Claude Original	Data extraction	Data extraction with different versions of Claude and specific prompts

AI, artificial intelligence; API, application programming interfaces; LLM, large language model; LaMDA, Language Model for Dialogue Applications; MeSH, Medical Subject Headings; PICO, Population Intervention, Control Outcome; PLS, plain language summary; qual., qualitative; quant., quantitative; RoB, risk of bias assessment; SR, systematic review.

[a] Preprint.
[b] Journal article.
[c] Communications (comments/letters/editorials/conference proceedings).
[d] Location of the primary institution of the first author.
[e] Version not specified.

For half of the studies (n = 20, 54%), authors drew a promising conclusion for LLM application in SR conduction, while about one-quarter of authors concluded neutrally (n = 9, 24%), and 22% (n = 8) of our 37 reports on LLMs discussed LLM use as nonpromising. For example, study prioritization in abstract screening was simplified using a question-answer framework in GPT, and showed higher precision and a substantial increase in efficiency, compared to other zero-shot ranking and BERT-family models [31]. Of the reports of promising findings, screening (n = 11) and data extraction (n = 6) appeared most often. On the other hand, an approach to evaluate reliability of GPT for performing RoB assessment of randomized trials (RCTs) using the RoB 2.0-tool found only slight to fair agreement between GPT and human reviewers [32]. Supporting the literature search was the SR step that was most frequently reported as nonpromising (n = 8). If preprint articles had been left out, there would have been nine promising conclusions less, two less neutral and one less nonpromising conclusions. Authors' overall conclusion regarding applicability of LLMs for the respective SR step is summarized in Figure 3.

As for study design, the majority of studies were designed as validation studies (n = 21, 57%), such as data extraction by Claude compared to human data extraction with already published studies as reference [23], or described an exploratory approach of LLM use for facilitating specific SR steps (n = 10, 27%), like formulation of a PubMed search string with GPT [33]. Few authors reported on specific tools they had developed (n = 2), such as search engine ''Meta-Phill'': Based on Population Intervention, Control Outcome components, literature metadata can be saved to a repository on a

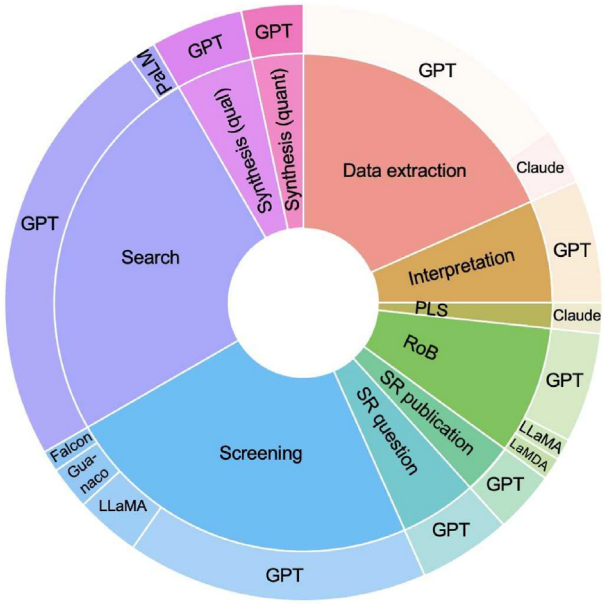


Figure 2. Pie-donut chart depicting proportions of systematic review (SR) steps (inner layer pie) and associated large language model (LLM) applications (outer layer donut). The inner layer represents the frequencies and proportions of all individual SR steps (n = 60) extracted across the 37 studies included (multiple counts per study were possible); the outer layer provides a breakdown of the percentage distribution of LLM types used for each SR step: Search (25% of all 60 individual SR steps, n = 15. GPT: 100%); Screening (23.3%, n = 14. GPT: 70%, LlaMA: 15%); Data extraction (18.3%, n = 11. GPT: 83.3%, Claude: 16.7%); risk of bias assessment (RoB) (8.33%, n = 5. GPT: 71.43%, LaMDA: 14.3%, LlaMA: 14.3%); Interpretation (6.7%, n = 4. GPT: 100%); SR question (5%, n = 3. GPT: 100%); Synthesis (qualitative) (5%, n = 3. GPT: 100%); Synthesis (quantitative) (3.3%, n = 2. GPT: 100%); SR publication (3.3%, n = 2. GPT: 100%); plain language summary (PLS) (1.7%, n = 1. Claude: 100%).

daily basis, technically supported by ChatGPT application programming interfaces [34]. Two studies were conceptualized as an SR on a medical question but supported by LLM, for example by asking GPT to write parts of the publication [35]. One report gave an opinion on the potentials of LLM application specifically for quality assessment and RoB evaluation in form of an editorial [36]. One article was a narrative review summarizing potential opportunities in information retrieval primarily with ChatGPT [37].

While 70% (n = 26) of studies were published in 2023, 30% (n = 11) were published in 2024, up to our search date on 26 February 2024. As for type of source, with the reported update in September 2024 on publication status of studies initially published as non−peer-reviewed preprint at the time of our search, 41% (n = 15) of the articles were peer-reviewed journal articles, 32% (n = 12) were comments, letters, editorials or conference papers, and 27% (n = 10) were non−peer-reviewed preprints. Most studies were carried out in the United States (n = 11, 30%), followed by the United Kingdom (n = 4, 11%) and Canada (n = 4, 11%).

4. Discussion

In this scoping review, we provide an overview of opportunities and limitations of LLMs to support SR conduct: We identified 37 articles on LLM applications, where systematic literature search, study selection (screening), and data extraction were most frequent, with OpenAI's GPT [11] being the most common LLM. Authors evaluated half of the applications as promising and half as neutral to nonpromising, highlighting both potential and limitations.

Building on a 2021 mapping review by Jimenez et al [4], who identified various ML applications in SR performance but no LLMs, we now observed that LLMs have indeed already found their way into many aspects of evidence synthesis: similar to our LLM findings, Jimenez et al identified screening, search, and data extraction as the most common ML-supported SR steps. Correspondingly, (semi) automated title abstract screening and literature search [38] as

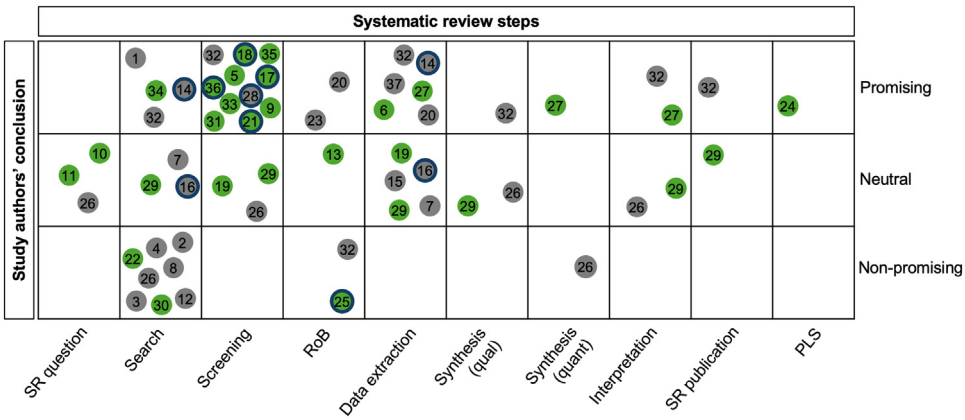


Figure 3. Bubble chart visualizing primary study design (green color: validation studies, gray color: other study designs, blue circle: preprint articles) and study authors' overall categorized conclusion (y-axis) for each systematic review (SR) step (x-axis). The categories (''promising'', ''neutral'', ''nonpromising'') reflect the study authors' own interpretation of results. Each bubble represents a study with study-ID as listed in Table 1. Studies evaluating several SR steps are represented multiple times accordingly.

well as data extraction [39] were the SR steps of greatest significance in further reviews on software and ML support in SRs. However, while the majority of publications included in Schmidt et al [39] describe natural language processing—supported data extraction from abstracts but rarely full texts, we now identified a report on LLM techniques that describes accurate extraction of data presented in text, figures or tables [23]. Contrariwise to training-intensive traditional ML, new LLM approaches require initial prompt engineering but no specific user training [23]. As such, LLMs demonstrated versatility across 10 out of 13 supported SR steps.

Despite the apparent technical simplicity of implementing LLMs in literature search, their evaluation—authors rated half of the approaches as nonpromising—shows limitations. In contrast, LLM support in study selection and data extraction appeared more favorable, with by far most (study selection) or at least a slight majority (data extraction) of the described application forms rated as promising, and the rest categorized as neutral.

We revealed notable gaps in LLM use for deduplication, full-text retrieval and evaluation of the certainty of evidence using the GRADE approach [40], underscoring constraints. This may be due to their complexity, technical challenges and hence omission of these steps in existing studies, or sufficiency of existing ML tools: Of note, traditional ML tools are already available for the three tasks, such as Deduklick [41] for deduplication, the review tool Covidence [30], which has recently gained some ability for full-text retrieval, and EvidenceGRADEr [42,43] for evaluating the certainty of evidence.

Regarding the use of LLMs to support SR tasks, there are a number of aspects to keep in mind. LLMs predict the next word based on previous words, resulting in coherent and fluent text output [14]. This creativity aids in tasks like conceiving SR ideas [44] or writing SR reports [35]. Conversely, unlike ML tools like RobotReviewer [45], LLMs do not rely on specialized algorithms and are not originally designed to process structured metadata or ensure transparent reproducibility, which is crucial for high-quality scientific literature [46] and evidence synthesis [47]. LLM output greatly varies with temperature settings and the exact wording of the prompt. Lower temperatures result in a more predictable output, higher temperatures enhance diversity in creation [48]. Moreover, minor prompt variations can alter the results [49]. Overall promising GPT-based title abstract screening lacked strict reproducibility even with zero temperature settings [50]. Efforts are underway to balance creativity and coherence in LLM outputs [51]. Capabilities vary depending on input length and session history [52,53]. Confabulations (“hallucinations”) [54] should be kept in mind. Further points of criticism include a lack of referencing appropriate and verifiable sources or retrieval of existing literature as well as a variability in nondeterministic LLM responses [21]. Cut-off dates for many LLM training datasets lead to incomplete information [33]. Specifically for validation studies, potential bias due to data contamination must be kept in mind, that is, if existing studies used as validation reference have already been included into an LLM’s training data set [55]. A way to counteract would be validation studies that use only the newest studies that cannot yet have been included into LLM training datasets. This, however, would significantly reduce the number of studies that can be taken for reference. Because of the authoritative-sounding but potentially inaccurate, incomplete, or biased outcome, some authors suggest application of LLM technology under human supervision and control only; further careful review and editing of the results by authors are needed [56]. The high proportion of validation studies suggests reliability as a basis for future usage, but also the authors’ urge for validation.

Focusing predominantly on GPT calls for caution as opaque decision-making and frequent updates may affect reproducibility [57] and objective and reliable (evidence) research. It could be beneficial to test other LLMs for their capabilities in SR production, also to determine possible differences between their performances. However, it must be kept in mind that test data can become outdated relatively quickly given the rapidly evolving field.

Our systematic search reveals a rapid increase in LLM-related publications, including a high proportion of non—peer-reviewed preprints, indicating immense interest in and potential impact for LLMs in SRs. While the sheer volume of papers on the topic may overwhelm peer review procedures, some reports lack sufficient data or methodology. For example, details on prompts were missing in articles [35], or ROBINS-I, conceptualized for RoB assessment of non-randomized studies of intervention, was used to assess RCTs with LLM assistance [58], where RoB 2.0 would have been the correct tool to test. More high-quality validation studies, conducted with strong methodological rigor, are needed for the most promising approaches.

Cierco Jimenez et al [4] suggest that ML- and software developers collaborate to improve already available applications. Similarly, our review can provide a basis for other researchers and app developers in the field.

4.1. Strengths and limitations

This scoping review provides a broad overview of current opportunities and pitfalls in using LLMs in SRs, with no similar work to date. Given the rapidly evolving nature of the field, our review may not be comprehensive, despite thorough searching. New performance assessments for LLM screening [22,59], data extraction [60], or RoB assessment [24] have emerged since our search. Additionally, we restricted inclusion to articles published in English or German, which means we may have missed articles published in other languages. While we aimed for optimal quality in elaborating authors’ overall conclusion as “promising”, “neutral” or “nonpromising”, some subjectivity regarding our categorization is inherent. Due to the

scoping review design, we did not conduct a critical appraisal of included articles. Therefore, the conclusions of the studies must be treated with some caution. To maintain the highest possible data quality, we followed a strict preregistered protocol and screened a large number of records including those on more traditional ML and BERT approaches. Most of these limitations are characteristic features of a scoping review, giving an orientation on the scope of a heterogeneous topic as a basis for future research.

4.2. Implications for research and practice

Future studies should improve transparency of reporting and ensure a rigorous methodology. Despite the outlined promising aspects, we emphasize the currently still numerous relevant limitations of LLM use, to preserve high quality and unbiased scientific research and evidence synthesis. Nonetheless, we assume that LLMs will play an increasingly important role in SR creation in the future—hopefully backed up with sound research.

5. Conclusion

This scoping review highlights the rapidly increasing role of LLMs in assisting SR conduction. A considerable number of studies showed promising results and drew optimistic conclusions. However, many approaches are currently exploratory in nature and not ready for direct transfer into research practice. LLMs should therefore still be used with caution and limited to specific SR tasks under human supervision.

Ethics statement

We used secondary data of full scientific articles published in journals or preprint servers. This article does not contain any studies with human or animal participants. Therefore, informed consent and ethical approval were not required.

CRediT authorship contribution statement

Judith-Lisa Lieberum: Writing − original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Markus Toews** Writing − review & editing, Methodology, Investigation, Data curation. **Maria-Inti Metzendorf:** Writing − review & editing, Investigation, Data curation. **Felix Heilmeyer:** Writing − review & editing, Investigation. **Waldemar Siemens:** Writing − review & editing, Conceptualization. **Christian Haverkamp:** Writing − review & editing, Conceptualization. **Daniel Böhringer:** Writing − review & editing, Visualization, Supervision, Resources, Methodology, Investigation, Conceptualization. **Joerg J. Meerpohl:** Writing − review & editing, Supervision, Resources, Methodology, Conceptualization. **Angelika Eisele-Metzger:** Writing − review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude 3.5 Sonnet to write and edit sections exclusively of the plain language summary. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

There are no competing interests for any author.

Acknowledgments

The authors would like to thank Kathrin Grummich for her advice in developing the draft search strategy, Philipp Kapp for his technical support in the use of Covidence software, Jacqueline Beck for supporting full-text retrieval.

Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclinepi.2025.111746.

Data availability

Our preregistered study protocol and extracted LLM and ML data as Excel spreadsheets can be found on OSF (Open Science Framework, https://osf.io/vdsgb). Supplement material includes our final search strategy and deviations from the preregistered study protocol.

References

[1] Mulrow CD. Rationale for systematic reviews. BMJ 1994;309:597−9.
[2] Marchevsky AM, Wick MR. Evidence-based pathology: systematic literature reviews as the basis for guidelines and best practices. Arch Pathol Lab Med 2015;139:394−9.
[3] Nussbaumer-Streit B, Sommer I, Hamel C, Devane D, Noel-Storr A, Puljak L, et al. Rapid reviews methods series: guidance on team considerations, study selection, data extraction and risk of bias assessment. BMJ Evid-Based Med 2023;28:418−723.
[4] Cierco Jimenez R, Lee T, Rosillo N, Cordova R, Cree IA, Gonzalez A, et al. Machine learning computational tools to assist the performance of systematic reviews: a mapping review. BMC Med Res Methodol 2022;22:322.

[5] Alchokr R, Borkar M, Thotadarya S, Saake G, Leich T. Supporting systematic literature reviews using deep-learning-based language models. Proceedings of the 1st International Workshop on Natural Language-based Software Engineering. Pittsburgh, Pennsylvania: Association for Computing Machinery; 2023:67−74.
[6] Bishop CM. Pattern Recognition and Machine Learning. New York, NY, USA: Springer US New York; 2006.
[7] Jordan MI, Mitchell TM. Machine learning: trends, perspectives, and prospects. Science 2015;349:255−60.
[8] Goodfellow I, Bengio Y, Courville A. Deep Learning. MIT Press; 2016. Available at: http://www.deeplearningbook.org. Accessed March 14, 2025.
[9] Janiesch C, Zschech P, Heinrich K. Machine learning and deep learning. Electron Markets 2021;31:685−95.
[10] Devlin J, Chang M-W, Lee K, Toutanova K. BERT: pre-training of deep bidirectional transformers for language understanding. arXiv 2019. North American Chapter of the Association for Computational Linguistics 10.48550/arXiv.1810.04805.
[11] OpenAI. Introducing ChatGPT. 2022. Available at: https://openai.com/index/chatgpt. Accessed March 14, 2025.
[12] Meta AI. Introducing LLaMA: A foundational, 65-billion-parameter large language model. 2023. Available at: https://ai.meta.com/blog/large-language-model-llama-meta-ai. Accessed March 14, 2025.
[13] Anthropic. Introducing Claude. 2023. Available at: https://www.anthropic.com/news/introducing-claude. Accessed March 14, 2025.
[14] Naveed H, Khan AU, Qiu S, Saqib M, Anwar S, Usman M, et al. A comprehensive overview of large language models. arXiv 2023. https://doi.org/10.48550/arXiv.2307.06435.
[15] Gehman S, Gururangan S, Sap M, Yejin C, Smith N. RealToxicityPrompts: Evaluating Neural Toxic Degeneration in Language Models. Findings of the Association for Computational Linguistics: EMNLP 2020. 3356−3369. Available at: https://aclanthology.org/2020.findings-emnlp.301. Accessed March 14, 2025.
[16] Weidinger L, Mellor JFJ, Rauh M, Griffin C, Uesato J, Huang P-S, et al. Ethical and social risks of harm from Language Models. arXiv 2021. https://doi.org/10.48550/arXiv.2112.04359.
[17] Lee P, Bubeck S, Petro J. Benefits, limits, and risks of GPT-4 as an AI chatbot for medicine. N Engl J Med 2023;388:1233−9.
[18] Lund BD, Wang T, Mannuru NR, Nie B, Shimray S, Wang Z. ChatGPT and a new academic reality: artificial Intelligence-written research papers and the ethics of the large language models in scholarly publishing. J Assoc Inf Sci Technol 2023;74:570−81.
[19] van de Schoot R, de Bruin J, Schram R, Zahedi P, de Boer J, Weijdema F, et al. An open source machine learning framework for efficient and transparent systematic reviews. Nat Mach Intell 2021;3:125−33.
[20] Distiller SR. DistillerSR Inc. 2021. Available at: https://www.distillersr.com. Accessed March 14, 2025.
[21] Qureshi R, Shaughnessy D, Gill KAR, Robinson KA, Li T, Agai E. Are ChatGPT and large language models "the answer" to bringing us closer to systematic review automation? Syst Rev 2023;12:72.
[22] Issaiy M, Ghanaati H, Kolahi S, Shakiba M, Jalali AH, Zarei D, et al. Methodological insights into ChatGPT's screening performance in systematic reviews. BMC Med Res Methodol 2024;24:78.
[23] Gartlehner G, Kahwati L, Hilscher R, Thomas I, Kugley S, Crotty K, et al. Data extraction for evidence synthesis using a large language model: a proof-of-concept study. Res Synth Methods 2024;15:576−89.
[24] Eisele-Metzger A, Lieberum J-L, Toews M, Siemens W, Heilmeyer F, Haverkamp C, et al. Exploring the potential of Claude 2 for risk of bias assessment: using a large language model to assess randomized controlled trials with RoB 2. medRxiv 2024. https://doi.org/10.1101/2024.07.16.24310483.
[25] Peters MDJ, Godfrey C, McInerney P, Munn Z, Tricco AC, Khalil H. In: Aromataris E, Munn Z, editors. Chapter 11: Scoping Reviews (2020 version). JBI Manual for Evidence Synthesis. JBI; 2020. Available at: https://synthesismanual.jbi.global. Accessed March 14, 2025.
[26] Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. Ann Intern Med 2018;169:467−73.
[27] Parisi V, Sutton A. The role of ChatGPT in developing systematic literature searches: an evidence summary. J EAHIL 2024;20:30−4.
[28] Bond M, Finnerty A, O'Mara-Eves A, O'Driscoll P, Thomas J, Minx J, et al. Digital evidence Synthesis tool evaluations. 2024. EPPI Visualiser database. Available at: https://eppi.ioe.ac.uk/eppi-vis/login/open?webdbid=435. Accessed March 14, 2025.
[29] Kung JY, Chojecki D. Filter to retrieve studies related to generative AI from the OVID MEDLINE database. Geoffrey & Robyn Sperber Health Sciences Library, University of Alberta; 2023; 2023. Available at: https://docs.google.com/document/d/1fPiY7GmI_Z4xZQTqZ58DySUTJousLlZr1GTCP-kDOLA/edit?tab=t.0#heading=h.ldbxqb34y1kj. Accessed March 14, 2025.
[30] Veritas Health Innovation. Covidence systematic review software. Melbourne, Australia. Available at: https://covidence.org. Accessed March 14, 2025.
[31] Akinseloyin O, Jiang X, Palade V. A novel question-answering framework for automated citation screening using large language models. medRxiv 2023. https://doi.org/10.1101/2023.12.17.23300102.
[32] Pitre T, Jassal T, Talukdar J, Shahab M, Ling M, Zeraatkar D. ChatGPT for assessing risk of bias of randomized trials using the RoB 2.0 tool: a methods study. medRxiv 2024. https://doi.org/10.1101/2023.11.19.23298727.
[33] Alaniz L, Vu C, Pfaff MJ. The utility of artificial intelligence for systematic reviews and boolean query formulation and translation. Plast 2023;11:e5339.
[34] Hatami N, Zarenezhad M, Javdani F, Keshavarz P, Kalani N, Sadeghinikoo A, et al. Meta-Phill: feasibility of a new metadata repository for Evidence-based practice in literature review. 2023. Research Square. Available at: https://www.researchsquare.com/article/rs-3133674/v1. Accessed March 14, 2025.
[35] Teperikidis E, Boulmpou A, Potoupni V, Kundu S, Singh B, Papadopoulos C. Does the long-term administration of proton pump inhibitors increase the risk of adverse cardiovascular outcomes? A ChatGPT powered umbrella review. Acta Cardiol 2023;78:980−8.
[36] Nashwan AJ, Jaradat JH. Streamlining systematic reviews: harnessing large language models for quality assessment and risk-of-bias evaluation. Cureus 2023;15:e43023.
[37] Huang Y, Huang J. Exploring ChatGPT for next-generation information retrieval: opportunities and challenges. ArXiv 2024. https://doi.org/10.48550/arXiv.2402.11203.
[38] Affengruber L, Nussbaumer-Streit B, Hamel C, Van der Maten M, Thomas J, Mavergames C, et al. Rapid review methods series: guidance on the use of supportive software. BMJ Evid Based Med 2024;29:264−71.
[39] Schmidt L, Olorisade BK, McGuinness LA, Thomas J, Higgins JPT. Data extraction methods for systematic review (semi)automation: a living systematic review. F1000Res 2021;10:401.
[40] Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. J Clin Epidemiol 2011;64:383−94.
[41] Borissov N, Haas Q, Minder B, Kopp-Heim D, von Gernler M, Janka H, et al. Reducing systematic review burden using Deduklick: a novel, automated, reliable, and explainable deduplication algorithm to foster medical research. Syst Rev 2022;11:172.
[42] Suster S, Baldwin T, Lau JH, Jimeno Yepes A, Martinez Iraola D, Otmakhova Y, et al. Automating quality assessment of medical evidence in systematic reviews: model development and validation study. J Med Internet Res 2023;25:e35568.
[43] Suster S, Baldwin T, Verspoor K. Analysis of predictive performance and reliability of classifiers for quality assessment of medical evidence revealed important variation by medical area. J Clin Epidemiol 2023;159:58−69.

[44] Gupta R, Park JB, Bisht C, Herzog I, Weisberger J, Chao J, et al. Expanding cosmetic plastic surgery research with ChatGPT. Aesthetic Surg J 2023;43:930−7.
[45] Marshall IJ, Kuiper J, Wallace BC. RobotReviewer: evaluation of a system for automatically assessing bias in clinical trials. J Am Med Inf Assoc 2015;23:193−201.
[46] Wallach JD, Boyack KW, Ioannidis JPA. Reproducible research practices, transparency, and open access data in the biomedical literature, 2015-2017. PLoS Biol 2018;16:e2006930.
[47] Page MJ, Moher D, Fidler FM, Higgins JPT, Brennan SE, Haddaway NR, et al. The REPRISE project: protocol for an evaluation of REProducibility and Replicability in Syntheses of Evidence. Syst Rev 2021;10:112.
[48] Democratizing Artificial Intelligence Research, Education, and Technologies (DARI.AI). Prompt engineering guide. Available at: https://dair.ai/projects/prompt-engineering. Accessed March 14, 2025.
[49] Salinas A, Morstatter F. The butterfly effect of altering prompts: how small changes and jailbreaks affect large language model performance. arXiv 2024. https://doi.org/10.48550/arXiv.2401.03729.
[50] Kataoka Y, So R, Banno M, Kumasawa J, Someko H, Taito S, et al. Development of meta-prompts for Large Language Models to screen titles and abstracts for diagnostic test accuracy reviews. medRxiv 2023. https://doi.org/10.1101/2023.10.31.23297818.
[51] Nguyen M, Baker A, Kirsch A, Neo C. Min P sampling: balancing creativity and coherence at high temperature. arXiv 2024. https://doi.org/10.48550/arXiv.2407.01082.
[52] Li D, Shao R, Xie A, Sheng Y, Zheng L, Gonzalez J, et al. How long can context length of open-source LLMs truly promise? NeurIPS 2023 Workshop on Instruction Tuning and Instruction Following. 2023. Available at: https://openreview.net/forum?id=LywifFNXV5. Accessed March 14, 2025.
[53] Wu Y, Hee MS, Hu Z, Lee RKW. LongGenBench: Benchmarking Long-Form Generation in Long Context LLMs. arXiv 2024. https://doi.org/10.48550/arXiv.2409.02076.
[54] McGowan A, Gui Y, Dobbs M, Shuster S, Cotter M, Selloni A, et al. ChatGPT and Bard exhibit spontaneous citation fabrication during psychiatry literature search. Psychiatry Res 2023;326:115334.
[55] Singh AK, Kocyigit MY, Poulton A, Esiobu D, Lomeli M, Szilvasy G, et al. Evaluation data contamination in LLMs: how do we measure it and (when) does it matter? arXiv 2024. https://doi.org/10.48550/arXiv.2411.03923.
[56] Anghelescu A, Firan FC, Onose G, Munteanu C, Trandafir AI, Ciobanu I, et al. PRISMA systematic literature review, including with meta-analysis vs. Chatbot/GPT (AI) regarding current scientific data on the main effects of the calf blood deproteinized hemoderivative medicine (actovegin) in ischemic stroke. Biomedicines 2023;11:02.
[57] Blum M. ChatGPT produces fabricated references and falsehoods when used for scientific literature search. J Card Fail 2023;29:1332−4.
[58] Mahuli SA, Rai A, Mahuli AV, Kumar A. Application ChatGPT in conducting systematic reviews and meta-analyses. Br Dent J 2023;235:90−2.
[59] Oami T, Okada Y, Nakada TA. Performance of a large Language Model in screening citations. JAMA Netw Open 2024;7:e2420496.
[60] Konet A, Thomas I, Gartlehner G, Kahwati L, Hilscher R, Kugley S, et al. Performance of two large language models for data extraction in evidence synthesis. Res Synth Methods 2024;15:818−24.